

Ornob Barua

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Portfolio link: [Ornob Barua | Mechanical Engineering Portfolio](#)

EDUCATION

B.S./M.S. Mechanical Engineering, University of California, Santa Barbara

B.S.: June 2027 M.S: June 2028

GPA: 3.84 Dean's Honors

Related Coursework: Mechatronics, Heat Transfer, Thermodynamics, Finite Element Analysis, Fluid Mechanics, Statics, Dynamical Systems, Machine Learning, MEMS fabrication, Vibrations, Heat Transfer, Design Optimization

EXPERIENCE

ASML

June 2026 - September 2026

Mechanical Engineer Intern

San Diego, California

- Developed angular arcmin tolerance stack up analysis for optical assembly to compare to requirement
- Established CMM workflow with MATLAB to conduct 3D tilt analysis to simulate predicted optical alignment.
- Created HTML angular tilt calculator to enable theoretical tilt measurement based on GD&T/real life assembly
- Developed technical documentation of all results and findings from analysis for future redesign of assembly
- Designed aluminum vee rail guide for CMM measurements, to reduce measurement uncertainty by ~5x

UCSB Nanofabrication Facility

March 2025 – March 2026

Processing Technician Intern

Goleta, California

- Operated scanning electron microscope (SEM) for imaging and analysis of different calibrations
- Performed FICP Si and SiO₂ calibrations using profilometry and ellipsometry to calculate selectivity and etch rate
- Executed photolithography using an ASML DUV stepper, followed by SEM analysis to verify pattern fidelity
- Performed ALD calibrations for Al₂O₃ and SiO₂ thin films, thickness verification with ellipsometry
- Completed calibrations for ICP-PECVD machine for SiO₂ and SiN thin films using metrology for analysis
- Improved maskless aligner (MLA) calibration process by redesigning calibration pattern with KLayout and Python, performing microscope and SEM-based analysis, and creating a traveler to guide new interns

Arista Networks

June 2025 – September 2025

Mechanical Engineer Intern

San Jose, California

- Designed and fabricated custom thermal test fixture for busbar power and return feeders
- Developed signal integrity test fixture for Time Domain Reflectometry measurements of line card and fabric module systems, enabling reliable debugging
- Utilized Creo Parametric and sheet metal design practices to model components effectively for test fixture
- Created detailed part and assembly engineering drawings and issued drawings to vendors for quotation
- Tested Arista 7800 installation guide through full chassis assembly to improve clarity and usability

Cadence Group, UCSB

October 2024 – June 2025

Undergraduate Research Assistant

Goleta, California

- Engineered a sensor system using an Arduino Nano and reed switch to quantify rod rotations in a reliable manner
- Designed a custom motor bracket and wood chassis with SolidWorks to ensure motor stability during operation
- Implemented electrical housing for the motor's power supply and sensor system to ensure safe operation

Projects

SiO₂/Glass/PDMS Microchannels

March 2026 – June 2026

- Developed and followed process followers utilizing thermal oxidation, wet etching, contact photolithography, ellipsometer, and profilometer to fabricate 3 different types of microchannels
- Analyzed data from microscopy, ellipsometry, and profilometry to fabricate chips to target dimensions
- Conducted current monitoring with electrode set up to test microchannels for microfluidic flow by applying 100V to 500V with channel filled with electrolyte solution and compare to analytical results

Automated Toilet Paper Replacement System

March 2026 – June 2026

- Engineered an electromechanical system integrating Arduino-based motor control, IR sensing, and custom laser-cut and 3D-printed components
- Optimized mechanism and control sequence to achieve automated replacement in ~ 4.73 seconds

SKILLS

Technical: Creo Parametric, SolidWorks, Fusion, GD&T, DFM, Cleanroom Manufacturing and Metrology, SEM, 3D printing, Arduino, MATLAB, COMSOL, Python, KLayout, Laser Cutting, Soldering, LaTeX, MS Excel

Professional: Presenting & Articulation, Teamwork, Time management, Adaptability, Problem Solving, Presenting & Articulation